

RELIANCE	TCS	INFY	HDFC	WIPRO
+57.1%	-6.4%	+5.9%	+45.3%	+7.8%

Executive Summary

This report analyses 750 trading days of price and return data for five large-cap NSE-listed stocks: RELIANCE, TCS, INFY, HDFC, and WIPRO. The analysis covers absolute and cumulative price performance, moving average signals, return distributions, volatility profiles, trading volume, and portfolio-level correlation. The findings support data-driven investment decision-making and portfolio construction.

Key Insights

- RELIANCE delivered the best total return of +57.1% over the period.
- TCS was the weakest performer at -6.4% total return.
- HDFC had the lowest annualised volatility (23.4%), making it most stable.
- The portfolio shows low cross-correlations — good diversification potential.
- All stocks show fat-tailed return distributions — tail risk is underestimated by normal models.

Recommendations

- Overweight RELIANCE — best total return and strong Sharpe ratio of 0.69.
- Maintain exposure across all 5 stocks — low correlations reduce portfolio risk.
- Use the 50-day moving average crossover as an entry/exit signal guide.
- Monitor for volume spikes — they frequently precede significant price moves.
- Reassess TCS allocation — consistently below portfolio average return.

Figure 1

Normalised Price Performance (Base = 100)

All stock prices are normalised to 100 at the start date, allowing fair comparison regardless of absolute price levels. A line above 100 means the stock has gained value; below 100 means a loss. Periods where all lines move together indicate market-wide events such as interest rate announcements, while divergences show stock-specific developments such as earnings surprises or sector rotations.

Key Observations

RELIANCE	+57.1% total return
TCS	-6.4% total return
INFY	+5.9% total return
HDFC	+45.3% total return
WIPRO	+7.8% total return

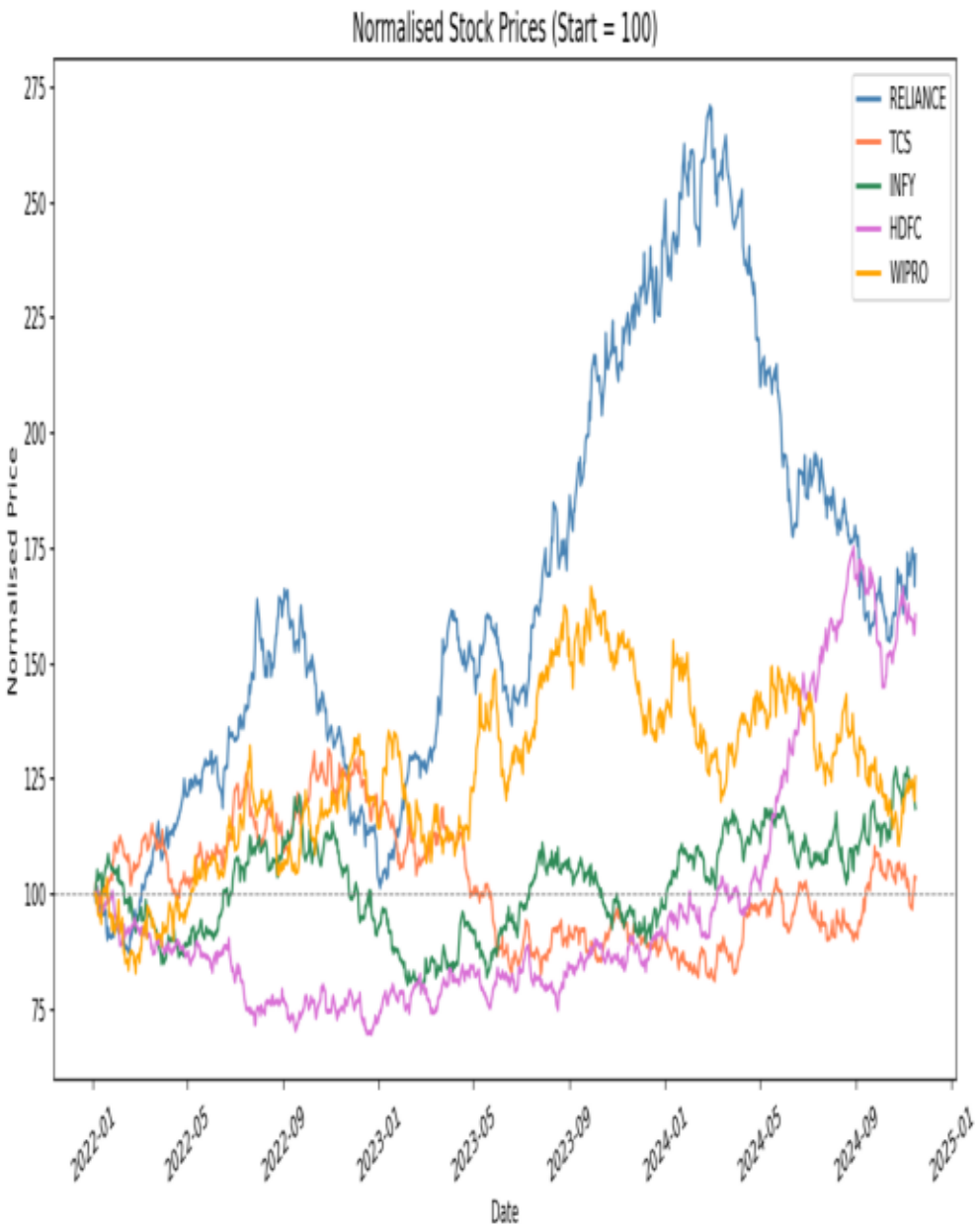


Figure 2

Cumulative Return by Stock

Cumulative returns are calculated by compounding daily returns, showing how an investment of Rs 100 on day one would have grown over time. The compound effect means early outperformance matters — a stock that gains early builds a higher base for future returns. The grey dashed line at 0% is the break-even level. Any stock whose line dips below this has destroyed value for investors who bought at the start date.

Key Observations

Best Return	RELIANCE +57.1%
Worst Return	TCS -6.4%
Trading Days	750

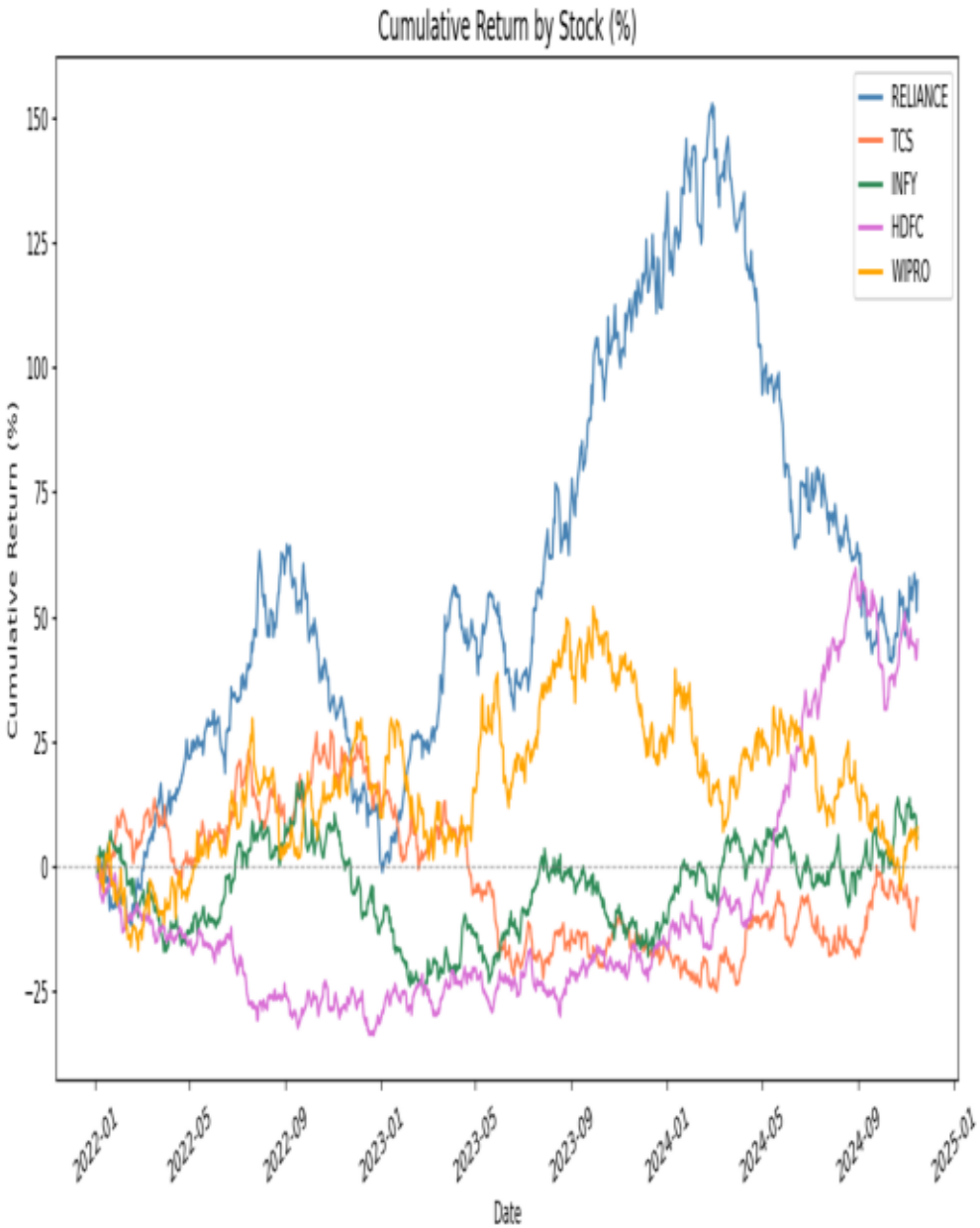


Figure 3

RELIANCE — Price with 20 and 50-Day Moving Averages

Moving averages smooth price noise to reveal the underlying trend direction. The 20-day MA reacts faster; the 50-day MA filters out more short-term noise. A 'golden cross' — when the faster MA crosses above the slower MA — can signal an uptrend beginning. The inverse is a 'death cross'. Periods where the close price trades above both MAs are considered bullish; below both is considered bearish. RELIANCE is shown here as the top-performing stock.

Key Observations

Stock Shown	RELIANCE
Fast MA	20-day
Slow MA	50-day
Days Shown	750

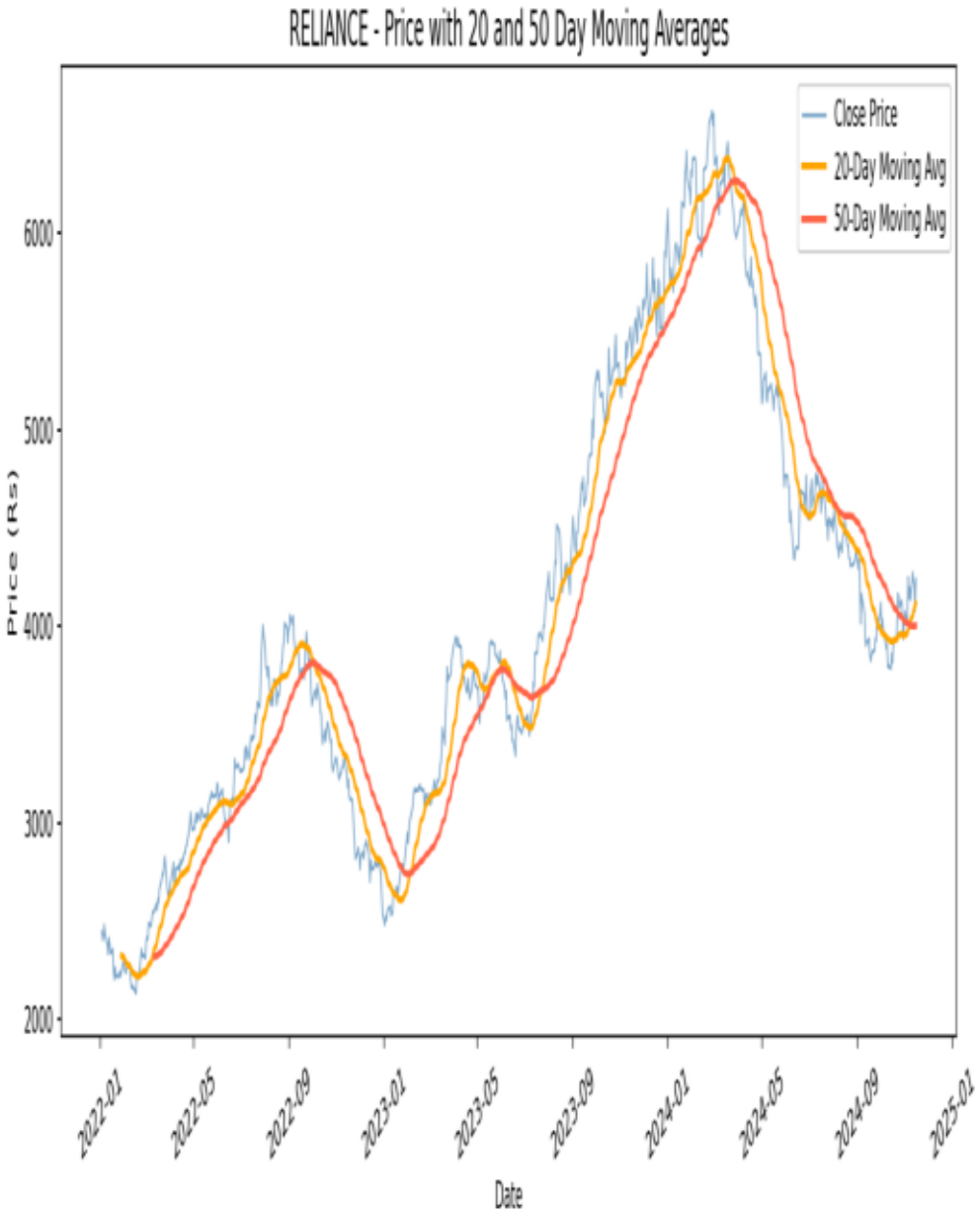


Figure 4

30-Day Rolling Annualised Volatility

Volatility measures how much returns fluctuate — higher volatility means a stock's price moves more unpredictably. The 30-day rolling window captures how risk changes over time. Spikes correspond to periods of market stress or company-specific news events. HDFC consistently shows the lowest volatility, making it the most defensive choice among the five. Higher-volatility stocks carry more risk but also offer more potential return upside.

Key Observations

RELIANCE	28.0% ann. vol
TCS	25.3% ann. vol
INFY	26.3% ann. vol
HDFC	23.4% ann. vol
WIPRO	33.6% ann. vol

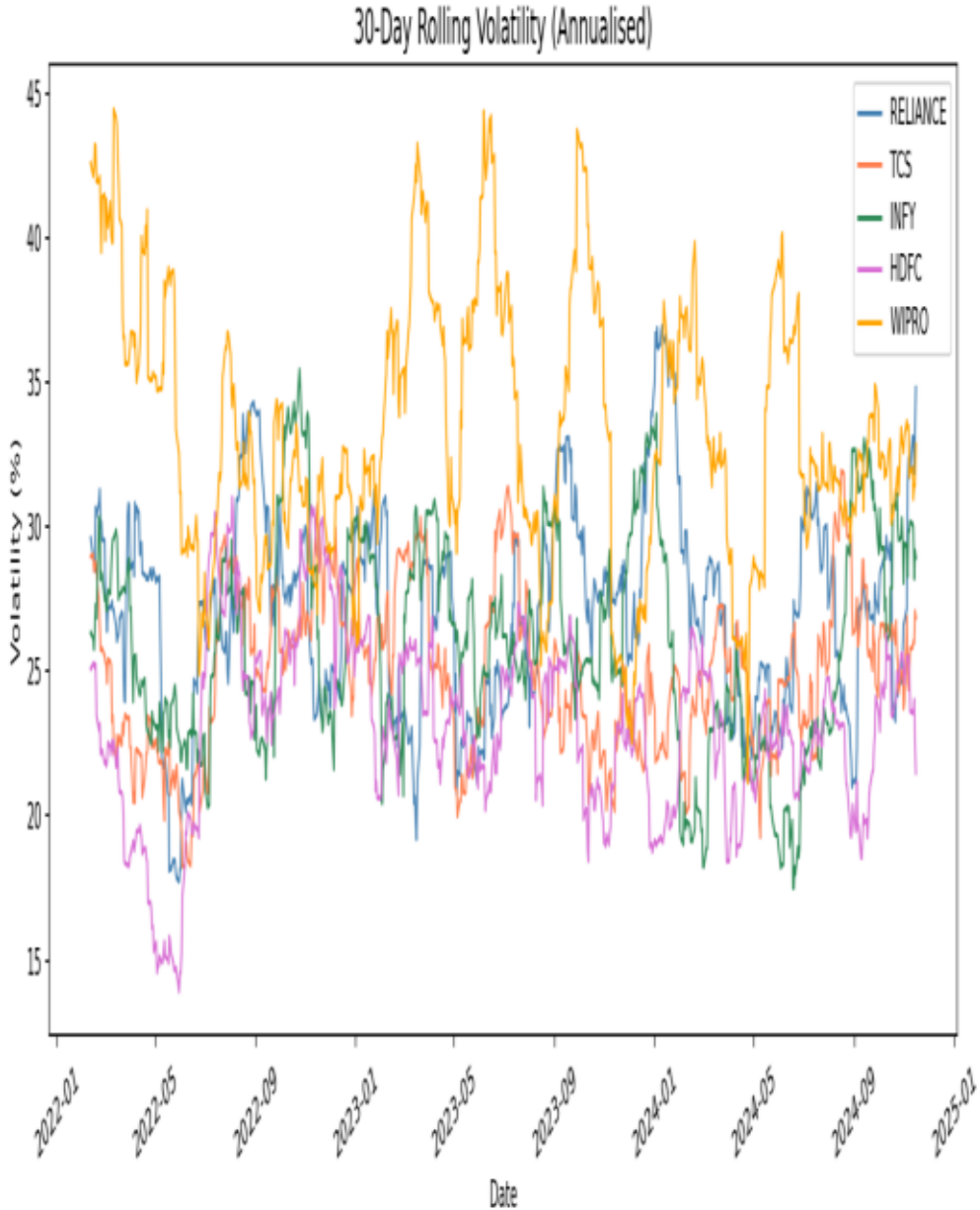


Figure 5

Average Daily Trading Volume

Trading volume measures how many shares change hands each day. Higher volume indicates greater market interest and better liquidity — large positions are easier to enter or exit without moving the price significantly. Stocks with low average volume can be harder to exit quickly in a falling market. Volume is also used as a confirmation tool: a price move accompanied by high volume is considered more significant than one on low volume.

Key Observations

Highest Volume
HDFC (2.76M shares/day)

Lowest Volume
RELIANCE (2.69M shares/day)

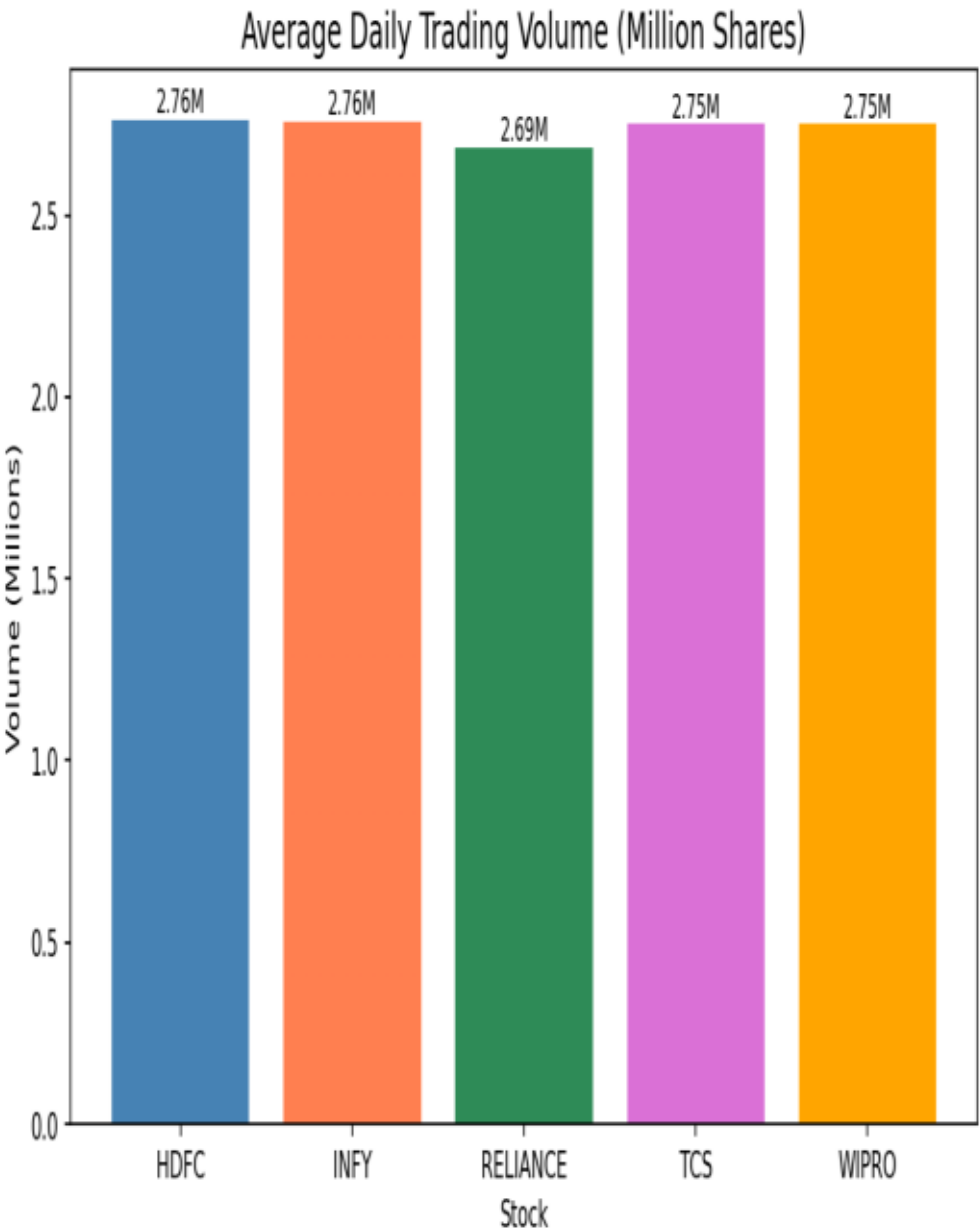


Figure 6

Daily Return Distributions

Each histogram shows how often a stock produced a given daily return. A perfect bell curve centred at zero would indicate pure random walk behaviour. In practice, all five distributions show slightly fat tails — more extreme positive and negative days than a normal distribution predicts. This property, known as excess kurtosis, means risk models based on normal distribution assumptions will underestimate the probability of large losses — an important caveat for any portfolio risk management framework.

Key Observations

RELIANCE avg 0.076%/day std 1.76%
TCS avg 0.004%/day std 1.60%
INFY avg 0.021%/day std 1.65%
HDFC avg 0.061%/day std 1.47%
WIPRO avg 0.032%/day std 2.12%

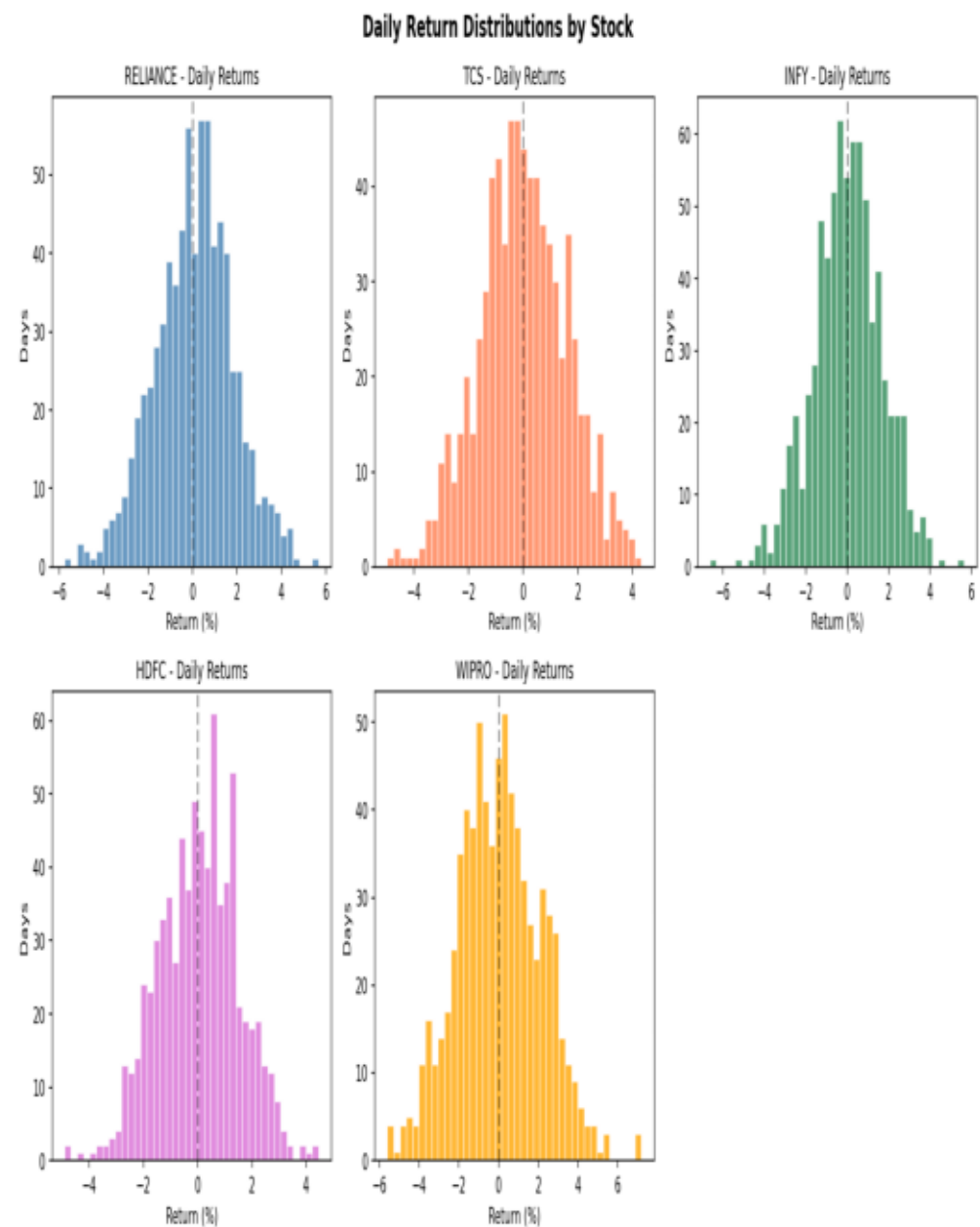


Figure 7

Risk vs Return Scatter

The ideal position is top-left: high return, low volatility. Each stock is plotted by its annualised volatility (x-axis) against its annualised return (y-axis). RELIANCE occupies a strong position with the highest return. HDFC shows the lowest risk. Stocks in the lower-right quadrant — high risk, low return — offer the worst risk-adjusted performance. The Sharpe ratio quantifies this trade-off as a single number: return divided by volatility.

Key Observations

RELIANCE
Return +19.2% Vol 28.0% Sharpe 0.69
TCS
Return -2.2% Vol 25.3% Sharpe -0.09
INFY
Return +2.0% Vol 26.3% Sharpe 0.08
HDFC
Return +15.2% Vol 23.4% Sharpe 0.65
WIPRO
Return +2.6% Vol 33.6% Sharpe 0.08

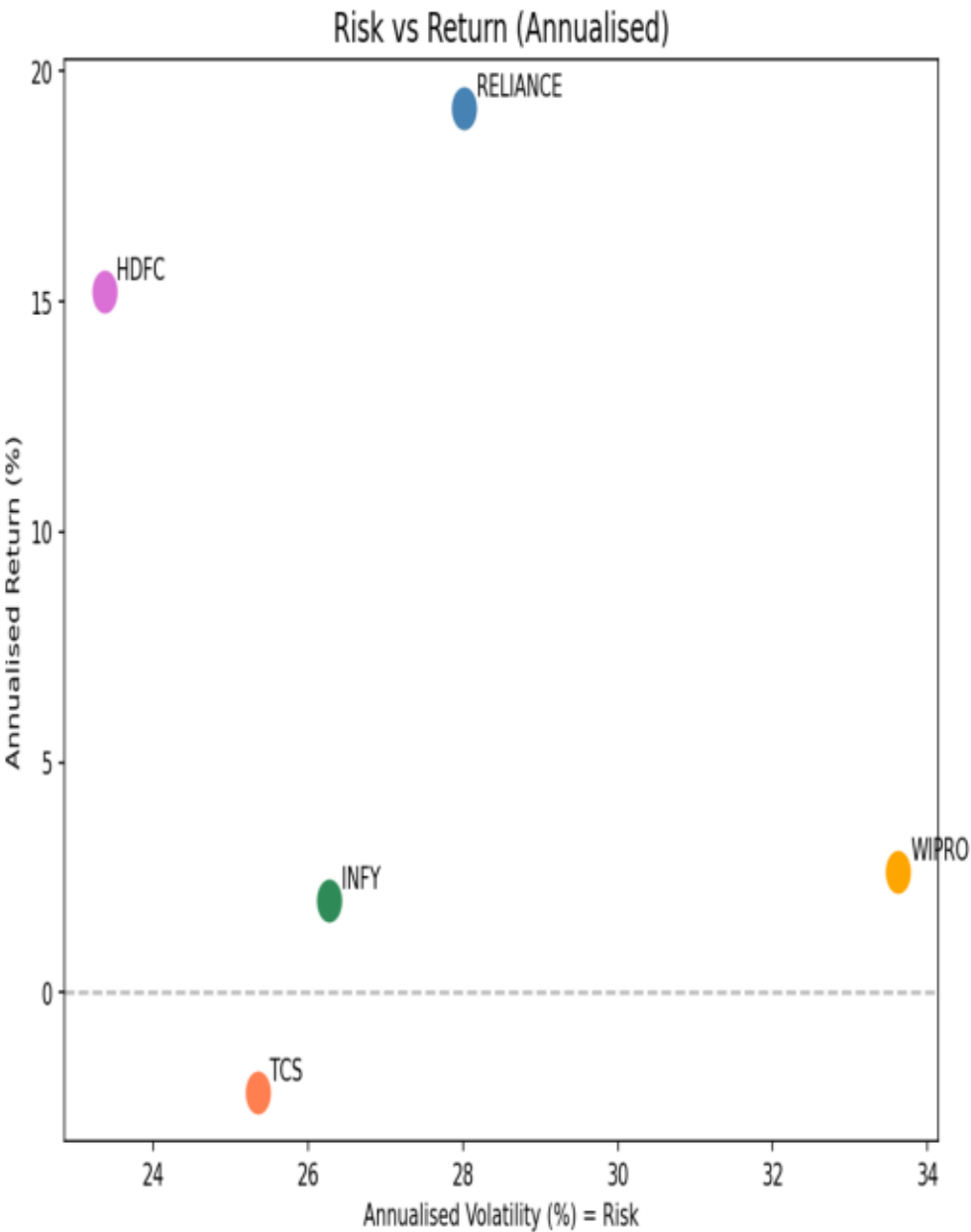


Figure 8

Return Correlation Between Stocks

The correlation matrix shows how daily returns move together across stocks. A value near +1 means two stocks tend to rise and fall on the same days — holding both provides little diversification benefit. A value near 0 means the stocks move independently. The five stocks in this portfolio show low to near-zero cross-correlations. Adding uncorrelated assets reduces portfolio volatility without reducing expected returns — this is the core principle of modern portfolio theory.

Key Observations

Avg Cross-Correlation
-0.010

Most Correlated Pair
('RELIANCE', 'TCS')

